

Chap-3

Long Multiplication:

$$\begin{array}{r}
 1000 \text{ ---} \rightarrow \text{Multiplicand} \\
 \times 1001 \text{ ---} \rightarrow \text{plier} \\
 \hline
 1000 \\
 00000 \\
 00000 \\
 10000 \dots
 \end{array}$$

multiplier right shift
 plier left shift

100010001 → product

Max length of prod = len of plicand + len of plier



prod = 128 bits



expand
 64 bits

128 bit → convert

* control test
 88888888 hardware
 88888888 connected.

how long multiplication happens?

1000×1001
 $\swarrow$ $\searrow$
 plicand plier

$10010 \times 11111 \rightarrow$ 5th iteration 2 20

check right, right shift

* 2 bit 5 bit num to multiply:

Multiplicand 20 $\rightarrow$ 10 bit
 plier $\rightarrow$ 10 bit

iteration	Multiplicand	Multiplicand	product
0	$1001 =$	$0000 \mid 1000$ MSB 8-bit (actual) LSB	0000 0000
1	0100 0100 (right shift)	$0001 \mid 0000$ (left shift)	0000 1000 plier 20 1 first time, plicand + product 20721
2	0010	0010 0000	0000 1000
3	0001	0100 0000	0000 1000
4	0000	1000 0000	0100 1000

$\swarrow$
 final ans.

* Multiplier or last 2 1's are used $\rightarrow$ multiplicand + product register
 Figure 1 shows the product register and the multiplicand register

also, then,
 multiplier's last 2 1's $\rightarrow$ shift multiplicand reg left 1 bit
 0's are used $\rightarrow$ shift multiplier reg right 1 bit

Intro to IEEE-754 floating point repres.

12 to store register X_{21} 2
 Addi $X_{21}, X_0, 12$

Floating point register store value $\rightarrow$ 32 bit

* 01 x 12 6 exponent
 4.5 x 10 $\rightarrow$ 31 23 16 9 2
 31 23 16 9 2
 31 23 16 9 2
 31 23 16 9 2

* single precision (32 bit)

Double	(64 bit)	48 bit
0 5	23 0	48 1
5 7	11	48 1

Normalized num:

1. dec / binary point at first only 1 digit
2. that digit must be non-zero

1 123.456 $\times$ $\rightarrow$ 1.0111 $\checkmark$

2 1.2345 $\checkmark$ $\rightarrow$ 0.11101 $\times$

3 0.123 $\times$ $\rightarrow$ 1101.1000 $\times$

* How to normalize a num:

* 1234.5678 $\rightarrow$ 112.54 $\times 10^3$

~~2122~~ $\rightarrow$ 1.1254 $\times 10^3$

* IEEE floating point format:

	sign bit	Exponent	Fraction
single p.	1 bit	8 bits	23
double p.	1 bit	11 "	52

sign bit = 0 $\rightarrow$ pos num
 = 1 $\rightarrow$ neg num

single precision: (32 bit)



Exponent = will be represented as unsigned num

8 bit unsigned binary range = 0 to $2^8 - 1$

$$= 0 \text{ to } 255$$

biased exponent range is 1 to 254

Biased exponent range = 0 to $2^8 - 1$

$$= 1 - 255 = 1 - 254$$

$$\text{Bias} = 2^{8-1} = 127$$

actual exponent range = (1 - 127 to 254 - 127)

$$= -126 \text{ to } 127$$



* Bias, Biased exponent calculation with exmpl.

$$\text{Bias} = 2^{n-1} - 1$$

size of field

Biased exponent = Actual exponent + Bias

Ex: 1.1011×2^3 , find biased exponent of given num in IEEE 754 single prec. format.

Soln:

1. check if num is normal or not:

$$1.1011 \times 2^3 \quad \text{actual} = 1.1011 \times 2^{\text{expo}}$$

$$\text{Biased exp} = 3 + 127 = 130 \text{ (Ans)}$$

* Single prec. so, Exponent field size = 8 bits

$$\text{bias} = 2^{8-1} - 1 = 2^7 - 1 = 127$$

Ex: 1.1011×2^{-8} , find biased expo of given num in IEEE single precision format where size of fraction field is 8 bits.

sign bit	expo	frac
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$$1 + x + 8 = 15$$

$$\therefore x = 6$$

$$\therefore \text{expo} = 6$$

$$\therefore \text{Bias} = 2^{6-1} - 1 = 2^{5-1} = 2^4 = 16$$

$$111011 \times 2^{-8}$$

0111	1011	1001	0010	...
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actual expon

$$\text{biased expon} = -7 + 16 = 9$$

Decimal to Floating point conversion:

1. Convert decimal num to binary
2. Normalize binary num
3. Find biased expon
4. Sign bit
5. Find fraction

Q. Convert 50.6749 to 32 bit IEEE-754 Floating point rep.

$$50 = 110010 \quad 1 = (0-) + 1 =$$

$$.6749 = 1010110011 \dots$$

$$50.6749 = 110010.1010110011 \dots \quad 1 - \Sigma = 2013$$

$$= 1.10010101010110011 \dots \times 2^5$$

actual exp

$$\text{Bias} = 2^{n-1} - 1 = 2^{8-1} - 1 = 127$$

single prec

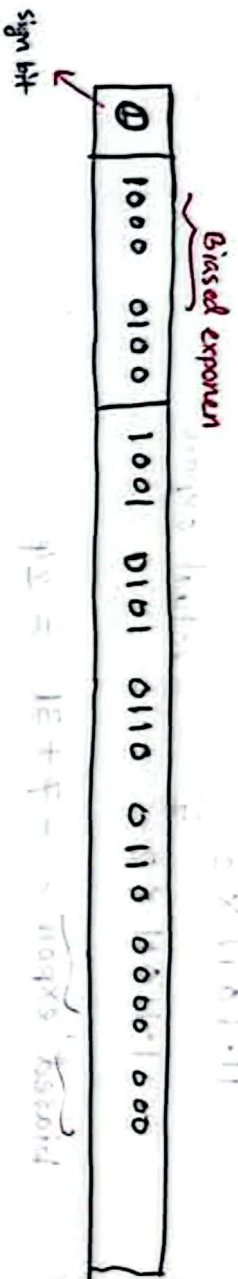
$$\therefore \text{Biased expon} = 127 + 5 = 132$$

1	00000000	10010101010110011
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= 10000100

sign bit = 0

Fraction = 1001 0101 0110 0110 0000 0000



→ 0x 424 A B 300

Ex-2: convert -0.0232 to 12 bit IEEE 754 floating point rep.

-0.0232 = -0.000000010

⇒ -0.0000010 × 2⁶

⇒ 1.0 × 2⁻⁶

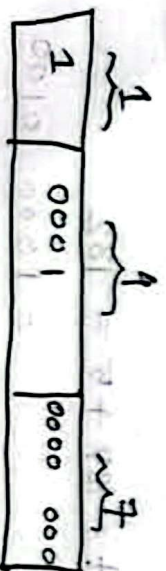
Biased exponent = Bias + actual exponent
= 7 + (-6) = 1

∴ Bias = 2ⁿ⁻¹

= 2³ - 1 = 7

sign bit = 1

Fraction = 0000 0000



$$\rightarrow 1000 \ 1000 \ 0000$$

$$= 0x880$$

(here dec)

Floating point to decimal Conversion:

Hex to binary

- i) Arrange binary according to format
- ii) Determine sign
- iv) Find out exponent from biased exponent
- v) Convert fraction to decimal
- vi) Decimal num = $(-1)^{\text{sign bit}} \times (1 + \text{fraction}) \times 2^{(\text{exponent})}$

Exmp:

0xF2400120, convert this single prec floating point num to decimal.

$\rightarrow$ 0xF2400120

i) 1111 0010 0100 0000 0000 0001 0010 0000

ii)

1	1110 0100	1000 0000 0000 0010 0100 0000
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1 8 23

sign Expon fraction

$$\text{iii) Biased expon} = 1110\ 0100 = 228$$

$$\text{Bias} = 2^{n-1} - 1 = 2^7 - 1 = 127$$

$$\text{Actual expon} = 228 - 127 = 101$$

$$\text{iv) Fraction} = 100\ 0000\ 0000\ 0001\ 0010\ 0000$$

$$= \left(\frac{1001000000000000000100100000}{2^{23}} \right)_2$$

$$= 0.500034\ 3323$$

$$\text{v) Decimal num} = (-1)^{\text{sign bit}} \times (1 + \text{frac}) \times 2^{\text{expon}}$$

$$= (-1)^1 \times (1 + 0.500034\ 3323) \times 2^{101}$$

$$= -1.500034\ 3323 \times 2^{101}$$

$$= -3.80303884 \times 10^{30} \quad (10^9 \text{ convert to decimal})$$

* round it upto 6 decimal points $\rightarrow$

$$-3.803039 \times 10^{30}$$

Floating point Addition/Subtraction

A & B both are floating point num

→ A + B (make sure, num is in binary)

i) normalize both A & B

ii) Align the bin point so that lower expon match with higher expon.

iii) Now add/sub accordingly.

iv) Normalise result, round if necessary

v) overflow/underflow check.

ex:

$$9.999 \times 10^1 + 1.610 \times 10^{-14}$$

$$= (99.99 + 0.1610)_{10}$$

$$= (100011.1111101101)_{2} + (0.0010100100100)_{2}$$

$$= 1.1000111111110101 \times 2^6 + 1.0100100100 \times 2^{-3}$$

(normalise)

$$= 1.1000111111110101 \times 2^6 + 0.000000001010100 \times 2^6$$

$$= 2^6 \times (1.1000111111110101 + 0.000000001010100)$$

$$= 1.10010 \times 2^6$$



Floating point Multiplication:

A & B both are floating point num.

⇒ A + B (Make sure the num is in binary)

i) Normalise both A & B

ii) Add exponents

iii) multiply accordingly

iv) Normalize result / round if necessary

v) Determine the sign from operation

$$\ominus(A) \times \ominus(B) = AB$$

Ex:

$$1.110 \times 2^5 \times 111.11 \times 2^{-7}$$

$$= 1.110 \times 2^5 \times 1.1111 \times 2^{-5} \quad (\text{normalise})$$

$$= (1.110 \times 1.1111) \times 2^{5+(-5)}$$

$= 10$

← calc A, (1110 × 11111)

7th digit
no. point

$$= 110110010$$

← (normalise)
product

$$= 11.0110010 \times 2^0$$

$$= 1.10110010 \times 2^1$$

यदि अलग अलग,

$$\begin{aligned} & \ominus (1.110 \times 2^5) \times (111.11 \times 2^{-7}) \\ & \Rightarrow 1.110 \times 2^5 \times 111.11 \times 2^{-7} \end{aligned}$$

यहाँ पूर्णांक और दशमलव भाग

$$\Rightarrow -1.1011 \times$$

* overflow / underflow :

range = (x to y)
 $\swarrow$ lower limit $\searrow$ upper limit

a is a number.

if $(x < a < y)$:

No overflow / underflow

if $(a < x)$:

underflow

if $y > a$:

overflow

Q: 18 bit. fraction bit $\rightarrow$ 12, bit.

check overflow/underflow.

Ans:



$$10) (0.000101 \times 2^{-85}) \times 2^{(10 \times 2^{-90})}$$

$$\Rightarrow 1.01 \times 2^{-85-4} \times 1.01 \times 2^{-90+1}$$

$$= 1.01 \times 2^{-89} \times 1.01 \times 2^{-89}$$

$$= (1.01 \times 1.01) \times 2^{-89+(-89)}$$

$$= (1.01 \times 1.01) \times 2^{-178}$$

$$= 1.1001 \times 2^{-178} \quad (\text{normalised})$$

$\rightarrow$ actual
expon

Ans:

(exponent - bias)

$$-0.75 = (-1)^s \times (1 + \text{fraction}) \times 2$$

$$\Rightarrow (-1)^1 \times (1 + .1)_2 \times 2$$